

Economic Design Optimization of Foundations

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Abstract: A geotechnical foundation design should address at least three basic requirements: ultimate limit state (ULS), serviceability limit state (SLS), and economics. Most conventional design approaches focus on ULS and/or SLS optimization, with economics being evaluated afterwards. As an alternative, this paper develops a design approach that explicitly considers the construction economics and results in a foundation that has the minimum construction cost. This design approach is expressed as an optimization process, in which the objective is to minimize construction cost, with the design parameters and design requirements as the optimization variables and constraints, respectively. This design approach is illustrated using a spread footing example. Because construction costs vary by locale, the economically optimized designs differ regionally. Sensitivity studies on soil properties and design requirements show that, for typical spread footing designs in cohesionless soils, Young's modulus (E) and the effective friction angle (ϕ') are the key parameters. A quantitative assessment illustrates the importance of soil property variability on cost. It is also found that, for typical spread footing designs, a relatively stringent ULS requirement generally ensures fulfillment of the SLS requirement.

DOI: 10.1061/(ASCE)1090-0241(2008)134:8(1097)

CE Database subject headings: Economic factors; Cost estimates; Optimization; Spread foundations; Limit states; Serviceability.

Introduction

A foundation design should address at least three basic requirements: ultimate limit state (ULS), serviceability limit state (SLS), and economics. The designed foundation must be safe from failure of its structural components and the surrounding geomaterial (i.e., ULS), and it must not result in objectionable deformation that could impair its usefulness or that of the supported structure (i.e., SLS). It is mandatory that designs should satisfy both the ULS and SLS requirements and, indeed, fulfillment of these requirements separates acceptable designs from unacceptable ones. However, as pointed out by Wellington (1887), the Father of Engineering Economy, the essentials of good engineering designs hinge on the third requirement, economics. His poignant quote is given below:

"To define it rudely but not inaptly, engineering is the art of doing that well with one dollar which any bungler can do with two after a fashion."

The significance of economics in geotechnical engineering designs is well recognized and is discussed in various textbooks (e.g., Lambe and Whitman 1969; Sowers 1979; and Coduto 1994). However, most of the discussions are at the conceptual level. We are not aware of any general analysis framework that

permits explicit consideration of ULS, SLS, and economics requirements in a consistent manner. Instead, the ULS and SLS requirements usually are addressed using a trial-and-error approach, in which a trial design is proposed and is checked against the ULS and SLS requirements, which is followed by revision of the trial design, if necessary. Economic considerations usually follow the technical and carry a lesser weight. Generally speaking, multiple designs could satisfy both the ULS and SLS requirements and, ideally, the final design should be the one with minimum construction cost. However, if the economics are not considered explicitly in the design, there is no guarantee that the final design is optimized economically.

This paper presents an approach that explicitly accounts for the ULS, SLS, and economics requirements within a consistent framework and ensures an economically optimized design. The geotechnical design process is framed as an optimization process, in which the optimization variables are the design parameters, the optimization objective is to minimize the construction costs, and the ULS and SLS requirements are treated as design constraints. This approach leads to an economically optimized design that satisfies both the ULS and SLS requirements. In this paper, the optimization framework is developed first, followed by a review of relevant construction cost estimates. Then, an example is described that applies the framework to a spread footing design, arguably the simplest design in geotechnical engineering. Finally, sensitivity studies are performed to explore the effects of design locale, soil properties, and ULS and SLS design requirements.

Conceptual Framework

Design is a process of determining a set of parameters that describe and specify a product that satisfies a series of performance requirements measured by their respective performance indices. In geotechnical engineering design, the product is a geo-structure, and the design needs to specify the type of geo-structure (e.g., spread footing or pile) and its associated details, such as dimen-

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Note. Discussion open until January 1, 2009. Separate discussions must be submitted for individual papers. To extend the closing date by one month, a written request must be filed with the ASCE Managing Editor. The manuscript for this paper was submitted for review and possible publication on January 26, 2007; approved on October 29, 2007. This paper is part of the *Journal of Geotechnical and Geoenvironmental Engineering*, Vol. 134, No. 8, August 1, 2008. ©ASCE, ISSN 1090-0241/2008/8-1097-1105/\$25.00.

sions and materials used. The performance requirements include, but are not limited to, safety and reliability or, more specifically, the ULS and SLS requirements, which are given by multiple performance indices, such as factor of safety, reliability index, and allowable deformation. Successful designs must satisfy all performance criteria.

Every product (or geo-structure) has a cost, which varies as the design parameters (e.g., materials and dimensions) change. The economics requirement of foundation design is to satisfy the mandatory performance requirements with minimum construction cost, and it can be addressed explicitly in design by optimization, in which minimizing the construction cost (Z) of the foundation is the optimization objective. Note that Z is a function of the design parameters ($x_1, x_2, x_3, \dots, x_n$) and other constants ($c_1, c_2, c_3, \dots, c_n$) that define the unit costs of construction materials, labor, and so on. The design parameters ($x_1, x_2, x_3, \dots, x_n$) are considered to be optimization variables, and the performance requirements, measured by performance indices ($PI_1, PI_2, PI_3, \dots, PI_n$), are treated as optimization constraints. The optimization is to obtain a set of design parameters ($x_1, x_2, x_3, \dots, x_n$) that result in a set of performance values ($PI_1, PI_2, PI_3, \dots, PI_n$) that are equal to or exceed the required performance indices ($PI_{1r}, PI_{2r}, PI_{3r}, \dots, PI_{nr}$) and also give the minimum Z . The optimization process can be expressed in a standard nonlinear programming format (Castillo et al. 2002) as follows:

Objective

$$\text{Minimize cost } Z = f(x_1, x_2, x_3, \dots, x_n, c_1, c_2, c_3, \dots, c_n)$$

Constraints

Performance Index 1, $PI_1(x_1, x_2, x_3, \dots, x_n)$, is equal to or exceeds PI_{1r}

Performance Index 2, $PI_2(x_1, x_2, x_3, \dots, x_n)$, is equal to or exceeds PI_{2r}

Performance Index 3, $PI_3(x_1, x_2, x_3, \dots, x_n)$, is equal to or exceeds PI_{3r}

⋮

Performance Index n , $PI_n(x_1, x_2, x_3, \dots, x_n)$, is equal to or exceeds PI_{nr}

Practical Constraints, $x_{il} \leq x_i \leq x_{iu}, i = 1, 2, 3, \dots, n$

in which x_{il} and x_{iu} = lower and upper bounds, respectively, for design parameter x_i and define the practical ranges of the design parameters. For example, consider a design parameter that defines the dimensions of a spread footing. These dimensions must be greater than zero, and therefore x_{il} must be greater than zero.

The economically optimized design then can be obtained by solving the optimization problem described above. There is substantial literature available on optimization, and well-established methodologies are applied widely for solving optimizations in various disciplines. Optimization software and toolkits also have been developed, such as the "Solver" function in Microsoft Excel. Background information on optimization is beyond the scope of this paper, and the reader is referred to Pedregal (2003) and Castillo et al. (2002) for details.

Cost Estimates

A key component in the optimization is calculating Z (cost estimate), which consists of two basic tasks: (1) determining material quantities and construction labor required (i.e., quantity takeoff); and (2) estimating reasonable costs for them (Means 1990). Essentials of cost estimates are reviewed in this section, while the reader is referred to Bledsoe (1992), Means (1990), and Waier (1993) for details.

There are several different levels (or types) of cost estimates, each of which has a different purpose and place in the construction estimating process. Means (1990) divides cost estimates into four different levels: (1) order of magnitude estimates; (2) square foot and cubic foot estimates (or SI equivalents); (3) assembly estimates; and (4) unit price estimates. The order of magnitude estimate could be described loosely as an educated guess and can be completed when only minimal information is available. One important purpose of this type of estimate is to provide information to an owner or client so that a budget-level decision can be made to proceed with the project.

The square foot and cubic foot estimate is an advanced preliminary estimate based on historical data. It is effective when features of the proposed project are known, but not yet designed (i.e., type of building or facility and proposed number of square feet or cubic feet).

The assembly estimate is an intermediate one prepared during the early stage of project design. An assembly refers to a collection of the individual tasks required to construct a composite work item. For example, consider a spread footing assembly, which usually includes five tasks: excavation, formwork, reinforcement, concrete, and compacted backfill. In these estimates, an assembly is treated as a cost unit, and the total cost is the product of the assembly unit cost and number of assemblies in the project.

The fourth level is the unit price estimate, in which the work is divided into the smallest possible work items, and a unit price is estimated for each item. The unit price then is multiplied by the quantity required to obtain the cost for the items, and the estimated cost is the sum of the costs for all items. The result is the most accurate and detailed estimate. All decisions regarding the construction materials and methods must have been made before the unit price estimates, because detailed working drawings, specifications, details on construction equipment, and productivity data are needed for determining the quantities of materials, equipment, and labor.

Clearly, each estimate level requires different amounts of time to complete and has different levels of accuracy. As the estimates progress from the least to the most accurate (i.e., from 1 to 4), the time needed for a modest building project might increase significantly from several minutes to several weeks. The required input data also vary significantly with estimate level. The estimator's experience on similar previous projects might be the major input data for the order of magnitude estimates, as opposed to the other three levels of estimates, which more frequently use published, annually updated, unit cost data, such as *Means Square Foot Costs* (Means 2006c), *Means Assemblies Cost Data* (Means 2006a), and *Means Building Construction Cost Data* (Means 2006b). Note that unit cost data are location dependent and vary from region to region. Costs for the same materials vary from region to region, or even from city to city, and labor rates and productivity also vary geographically. Therefore, it is not surprising to find that the unit costs in major cities (e.g., New York City) are much higher than those in rural areas.

For the best optimization, it would be desirable to adopt the level of cost estimates that are the most accurate, i.e., the unit price estimates. However, it is difficult to apply these estimates directly because they require detailed working drawings, specifications, details on construction equipment, and productivity data, none of which obviously are available during design. In this paper, a simplified unit price estimate is developed that uses the unit prices developed for assembly cost data (Means 2006a).

Table 1 shows an assembly example for constructing a

Table 1. Calculation of Spread Footing Assembly Unit Price (after Means 2006a)

Spread footing (0.91 m × 0.91 m × 0.30 m deep)	Quantity	Units	Cost per assembly United States dollars (USD)			Unit price (USD)
			Material	Labor	Total	
Excavation	0.451	m ³	N/A ^a	11.35	11.35	25.16
Formwork	1.115	m ²	7.80	50.16	57.96	51.97
Reinforcement	5.443	kg	5.61	6.15	11.76	2.16
Concrete	0.253	m ³	37.62	6.27	43.89	173.96
Compacted backfill	0.199	m ³	N/A ^a	0.79	0.79	3.97
Total	N/A ^a	N/A ^a	51.03	74.72	125.75	N/A ^a

Note: Cost data represent United States national average for 2007.

^aN/A=not applicable.

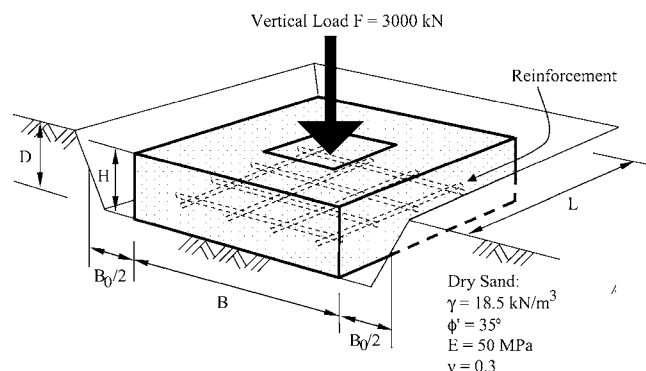
0.91 m × 0.91 m × 0.30 m deep spread footing. The first column contains the five individual tasks cited previously. The second and third columns show the quantities and associated units for these tasks. The United States national average costs for materials and labor are given in the fourth and fifth columns, respectively, and the sixth column shows the total costs. The unit prices are calculated by dividing the sixth column by the second. The quantities of the five individual tasks are functions of the design parameters (e.g., dimensions) and can be estimated. The following example illustrates this process for a spread footing.

Spread Footing Design

Spreading footing design is arguably among the simplest in geotechnical engineering, and therefore it is used to illustrate the optimization framework.

Design Example

Fig. 1 illustrates a spread footing in dry sand with a unit weight (γ) = 18.5 kN/m³ and an effective friction angle (ϕ') = 35°. The sand has a Young's modulus (E) = 50 MPa and a Poisson's ratio (ν) = 0.3. The groundwater is at considerable depth, so the spread footing is not affected by groundwater. The footing is designed to support a column that transfers a vertical load (F) = 3,000 kN to the footing. Three key design parameters for the spread footing are the depth (D), width (B), and length (L), as shown in Fig. 1.

**Fig. 1.** Spread footing design example

ULS and SLS Designs

The ULS and SLS requirements for the spread footing are given by the required factor of safety (FS_r) against soil failure and the allowable settlement (δ_r), respectively. In this example, $FS_r = 3$ and $\delta_r = 25$ mm are used as the design criteria. In addition, the foundation depth (D) should be greater than a minimum depth (D_1) = 0.5 m to prevent frost damage and should be limited to a maximum depth (D_u) = 2.0 m to minimize disturbance to adjacent structures.

The ultimate bearing capacity (q_{ult}) of a footing on a cohesionless, horizontal, sand deposit can be calculated from Eq. (1), for general shear and a concentrically applied vertical load (Vesic 1975)

$$q_{ult} = 0.5B\gamma'N_\gamma\zeta_{\gamma s}\zeta_{\gamma d} + q'N_q\zeta_{qs}\zeta_{qd} \quad (1)$$

in which γ' = effective unit weight of soil below footing; q' = effective overburden stress at foundation level; N_γ and N_q = bearing capacity factors, and $\zeta_{\gamma s}$, $\zeta_{\gamma d}$, ζ_{qs} , ζ_{qd} = shape and depth factors, as given below

$$N_\gamma \approx 2(N_q + 1)\tan\phi' \quad (2)$$

$$N_q = (e^{\pi \tan\phi'})\tan^2(45 + \phi'/2) \quad (3)$$

$$\zeta_{\gamma s} = 1 - 0.4(B/L) \quad (4)$$

$$\zeta_{\gamma d} = 1 \quad (5)$$

$$\zeta_{qs} = 1 + (B/L)\tan\phi' \quad (6)$$

$$\zeta_{qd} = 1 + 2\tan\phi'(1 - \sin\phi')^2[(\pi/180)\tan^{-1}(D/B)] \quad (7)$$

Then, the factor of safety (FS) can be calculated as

$$FS = \frac{q_{ult}}{F/BL} \quad (8)$$

For this example, the settlement (δ) is calculated using the following elastic solution (Poulos and Davis 1974)

$$\delta = \frac{F(1 - \nu^2)}{\beta_z E \sqrt{BL}} \quad (9)$$

in which β_z = shape factor given in Fig. 2 (Whitman and Richart 1967). Although this example uses an elastic solution to calculate δ , the economically optimized design framework is independent

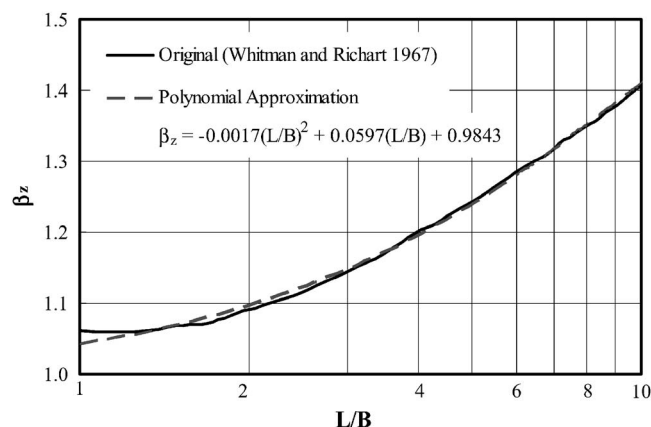


Fig. 2. Shape factor β_z

of the calculation model used for δ , and alternate nonelastic solutions can be used.

Conventional design begins with an initial assumption, followed by calculations, and then comparison of the calculated FS with FS_r and the calculated δ with δ_r . For example, a trial design with $D=0.6$ m and $B=L=2.6$ m is acceptable, because the calculated FS and δ are 3.1 and 20 mm, respectively, which satisfy the design requirements of $FS_r=3$ and $\delta_r=25$ mm. In general, many designs can be acceptable that satisfy both design requirements. For instance, another design with $D=2.0$ m, $B=1.7$ m, and $L=2.4$ m gives $FS=3.8$ and $\delta=25$ mm. Judgment and experience frequently are used to select a final design that reflects the economics issue. However, because economics are not addressed directly, there is no guarantee that the selected design yields the minimum construction cost.

Construction Cost Estimates

The cost of a spread footing can be estimated using the simplified unit price estimates, which require unit prices and quantities for each individual task. The quantities of the five tasks are a function of the design parameters and can be calculated explicitly. Excavation is measured by its volume, and this quantity (Q_e) can be calculated from Fig. 1 as

$$Q_e = (B + B_0)(L + L_0)D \quad (10)$$

in which B_0 and L_0 =overexcavation distances along the width and length directions, respectively. In foundation construction, overexcavation is necessary to facilitate placement of formwork and concreting. The overexcavation distances frequently are assumed as 0.3 m in cost estimates (Bledsoe 1992), so $B_0=L_0=0.3$ m is assumed herein.

Formwork is measured by its area, and this quantity (Q_f) can be calculated as

$$Q_f = 2H(B + L) \quad (11)$$

Concrete is measured by its volume, and this quantity (Q_c) can be calculated as

$$Q_c = BLH \quad (12)$$

in which H =height of footing. In this paper, $H=0.6$ m, which is typical for a spread footing resisting a concentric 3,000 kN vertical load. Reinforcement of spread footings is measured by its mass, and the quantity (Q_r) frequently is assumed proportional to

the concrete volume (Q_c) in cost estimates. Q_r can be expressed as

$$Q_r = kQ_c \quad (13)$$

in which k =proportionality coefficient that is taken as 29.67 kg/m³ (Means 1990) herein. Compacted backfill is measured by its volume, and this quantity (Q_b) is the difference between Q_e and Q_c , expressed as

$$Q_b = Q_e - Q_c \quad (14)$$

Then the construction cost (Z) of the spread footing can be estimated as

$$Z = Q_e c_e + Q_f c_f + Q_c c_c + Q_r c_r + Q_b c_b \quad (15)$$

in which c_e , c_f , c_c , c_r , and c_b =unit prices for excavation, formwork, concrete, reinforcement, and compacted backfill, respectively, as summarized in the last column of Table 1.

Economically Optimized Design

The economics requirement is considered explicitly in the spread footing design using the optimization framework, in which the optimization objective is to minimize Z , with the optimization variables B , L , and D . Both the ULS and SLS requirements are considered as optimization constraints and are reflected by FS and δ . In addition, B , L , and D are subjected to some practical constraints, i.e., B and L should be greater than zero, and D should be between 0.5 and 2.0 m to prevent frost damage or disturbance to adjacent structures. The optimization process can be expressed in a standard nonlinear programming format, as follows:

Objective

$$\text{Minimize cost } Z = f(B, L, D, H, B_0, L_0, k, c_e, c_f, c_c, c_r, c_b)$$

Constraints

$$\text{ULS requirement, } FS \geq 3$$

$$\text{SLS requirement, } \delta \leq 25 \text{ mm}$$

$$B > 0, L > 0, \text{ and } 2.0 \text{ m} \geq D \geq 0.5 \text{ m}$$

Then Z , FS, and δ are calculated using Eqs. (15), (8), and (9), respectively. To calculate β_z in Eq. (9), the β_z versus L/B relationship in Fig. 2 is approximated by the following second-order polynomial function:

$$\beta_z = -0.0017(L/B)^2 + 0.0597(L/B) + 0.9843 \quad (16)$$

The optimization model is set up in a Microsoft Excel spreadsheet and is solved using the Excel function Solver. Details are given in the Appendix. The minimum Z is USD 1,086, which occurs when $D=1.38$ m, $B=1.86$ m, and $L=2.30$ m. Table 2 summarizes details of the optimized design and comparison with two other design alternates described previously in the section "ULS and SLS Designs." Although the three designs satisfy both the ULS and SLS requirements, they have different values of B , L , and D , which results in different construction costs. Since the economics are not considered explicitly in conventional designs, the construction cost of the conventional footing might be significantly higher than the cost of the optimized design. Table 2 shows that the cost of example 1 is 31% higher than the optimized design. In contrast, the optimization process includes the construction cost estimates and ensures that the final design can be optimized with minimum construction cost.

Table 2. Comparison of Three Spread Footing Designs

Design option	Width, <i>B</i> (m)	Length, <i>L</i> (m)	Depth, <i>D</i> (m)	Cost (USD)	Difference (%)
Optimized	1.86	2.30	1.38	1086	N/A ^a
Example 1	2.60	2.60	0.60	1421	31
Example 2	1.70	2.40	2.00	1143	5

^aN/A=not applicable.

It should be noted that the construction cost also is affected by other factors, such as constructability, construction schedule, local contractor's preference, and availability of standard or local materials (e.g., formwork sizes). Although these factors are not included in this example, the optimization framework can be extended readily to incorporate these factors as optimization constraints.

Sensitivity Study

Effect of Locale

Construction costs vary from region to region, or even from city to city, as materials costs, labor rates, and productivity vary geographically. Because the construction costs are considered explicitly in the design, the location of the designed geo-structures and the geographic variation of construction costs can result in different optimized designs.

In cost estimates, the location effect is addressed using the city cost index. This index for the United States national average is taken as 100, and the indices for 316 major cities throughout the United States and Canada are given by Means (2006a). For example, consider the cost indices for New York City, which are 110.5 and 164.1 for materials and labor for foundation work, respectively. The materials and labor unit prices in New York City are adjusted by multiplying the ratio of the city cost indices (1.105 and 1.641) to the national average unit prices, such as the data in Table 1. Table 3 summarizes the calculation of spread footing unit prices in New York City. The fourth and fifth columns in Table 3 are obtained from the comparable columns in Table 1 and the city cost index ratios (1.105 and 1.641). The sixth column shows the total costs, and the seventh column summarizes the unit prices. Table 4 summarizes the unit prices of spread footings for the United States national average and New York City, which shows higher prices in New York City. Table 4 also in-

cludes the unit prices for Beijing, the capital of China, which are extracted from construction cost data in China (DLS 2006; CMC 1995).

The comparable, economically optimized designs of spread footings for the United States average, New York City, and Beijing are summarized in Table 5. The optimized design for New York City is a square footing with $B=L=2.10$ m and $D=1.25$ m. This design is slightly different from the one for the United States national average, with shallower depth (1.25 m versus 1.38 m), slightly larger footprint ($2.10 \times 2.10 = 4.41$ m² versus $1.86 \times 2.30 = 4.28$ m²), and a construction cost that is about 1.4 times higher than the United States national average cost. The optimized design for Beijing has $B=1.18$ m, $L=3.15$ m, and $D=2.00$ m, which is significantly different from the one for the United States national average. The depth ($D=2.00$ m) is the upper bound value and is significantly larger than the depth ($D=1.38$ m) for the United States national average. On the other hand, the footprint ($1.18 \times 3.15 = 3.72$ m²) is smaller than that for the United States national average ($1.86 \times 2.30 = 4.28$ m²).

All designs result in $FS=3$ and $\delta=25$ mm and reflect different strategies in fulfilling the design requirements. The footing might be either relatively shallow with a relatively large size or relatively deep with a relatively small size. The variation of foundation depth leads to different quantities of excavation and backfill, the construction costs of which are dominated by labor costs, as shown in Tables 1 and 3. In contrast, the variation of footing size results in different quantities of formwork, concrete, and reinforcement, the construction costs of which are much more affected by materials costs, as shown in Tables 1 and 3. Since construction cost is a product of quantity and unit price, the economically optimized design, which explicitly considers the construction cost, depends on the unit price of each individual work task and, particularly, the relative ratios between them. As the relative ratio changes, the economically optimized design might also change. Examination of Table 4 shows that, although all unit prices in Beijing are substantially smaller than those for the

Table 3. Unit Prices of Spread Footings in New York City

Spread footing (0.91 m $\times$ 0.91 m $\times$ 0.30 m deep)	Quantity	Units	Cost per assembly (USD)			Unit price (USD)
			Material	Labor	Total	
Excavation	0.451	m ³	N/A ^a	18.63	18.63	41.30
Formwork	1.115	m ²	8.62	82.31	90.93	81.55
Reinforcement	5.443	kg	6.20	10.09	16.29	2.99
Concrete	0.253	m ³	41.57	10.29	51.86	204.98
Compacted backfill	0.199	m ³	N/A ^a	1.30	1.30	6.51
Total	N/A ^a	N/A ^a	56.39	122.62	179.00	N/A ^a

Note: Calculated according to Table 1, using city cost indices of 110.5 and 164.1 for materials and labor, respectively.

^aN/A=not applicable.

Table 4. Summary of Unit Prices of Spread Footings for Three Locations

Work task	Units	Cost per unit (USD)		
		United States national average	New York City, United States	Beijing, China ^a
Excavation	m ³	25.16	41.30	1.38
Formwork	m ²	51.97	81.55	4.00
Reinforcement	kg	2.16	2.99	0.54
Concrete	m ³	173.96	204.98	47.50
Compacted backfill	m ³	3.97	6.51	0.22

^aExtracted from DLS (2006) and CMC (1995), assuming that the currency exchange rate between USD and Chinese currency, RMB is 1:8.

United States national average, work tasks that are dominated by labor, such as excavation and backfill, have significantly lower relative unit prices than those dominated by materials, such as reinforcement and concrete. Therefore, the optimized design in Beijing aims to use as much excavation and backfill as possible, resulting in an upper bound depth of 2 m with a relatively small footing size. In contrast, optimization of the design for the United States national average shows that a depth of 1.38 m with a relatively large footing size leads to the minimum construction cost. Similarly, as indicated by the materials and labor cost indices in New York City (110.5 and 164.1, respectively), labor cost increases more significantly than materials cost. Therefore, the optimized design in New York City aims to use less excavation and backfill, resulting in a relatively shallow depth of 1.25 m with a relatively large footing size. Because unit prices are geographically dependent, the economically optimized designs can vary by locale.

Although this paper focuses on the effect of locale, the concept is equally applicable to time. As time passes, materials cost, labor rates, and productivity change, resulting in possible alterations of the economically optimized design over time.

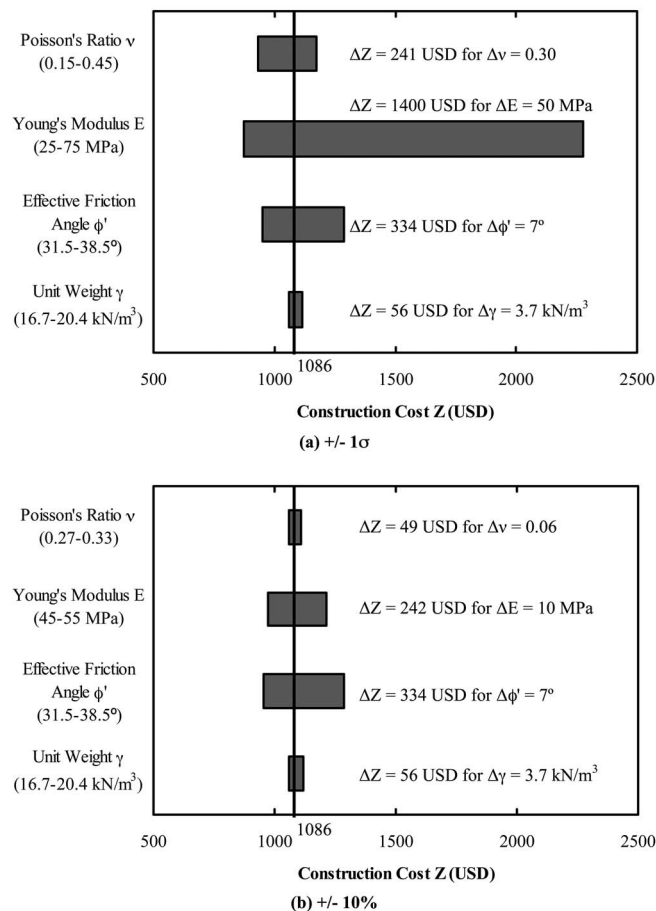
Effect of Soil Properties

Ground conditions and soil properties are key factors that control geotechnical engineering designs, and therefore proper site investigation is necessary to define the ground conditions and to establish design input parameters. Furthermore, site investigation should be conducted in a cost-effective manner. An economically optimized design permits quantitative assessment of the benefits of improving soil property characterization by means of a sensitivity study.

Soil properties for the spread footing design example include the soil unit weight (γ)=18.5 kN/m³, effective friction angle (ϕ')=35°, Young's modulus (E)=50 MPa, and Poisson's ratio (ν)=0.3. These values are taken as baseline values in the sensi-

Table 5. Summary of Optimized Designs for Three Locations

Optimized design	United States national average		
	United States national average	New York City, United States	Beijing, China
Width, B (m)	1.86	2.10	1.18
Length, L (m)	2.30	2.10	3.15
Depth, D (m)	1.38	1.25	2.00
Cost (USD)	1,086	1,512	179

**Fig. 3.** Sensitivity study on soil properties

tivity study, and variations of the soil properties are defined in accordance with their inherent variability. Phoon and Kulhawy (1999) characterized geotechnical variability and provided typical values of coefficient of variation (COV) for various soil parameters. The COVs of γ , ϕ' , and E are about 9, 5–15, and 15–65%, respectively. In this study, the variations of γ , ϕ' , and E are taken as their baseline values ± 1 SD, which are taken as 10% of their baseline values for γ and ϕ' and 50% of its baseline value for E . Since no data for ν are available, its variation is taken as the same as that for E .

Fig. 3(a) shows variations of construction costs for the economically optimized designs that result from variations of γ , ϕ' , E , and ν , compared to the baseline construction cost (USD 1,086). This figure shows that E has the most significant effect on the construction costs, followed by ϕ' . When E decreases to 25 MPa, the cost increases to about USD 2,300, more than twice the baseline value. The effects of ν and γ are relatively minor, and the variation of γ only leads to a change of construction costs of about 5%.

The most important soil parameter controlling the design is E , and therefore, when performing site investigations, attention should be directed to E . Note also that although the variation of E is symmetrical about the baseline value, the corresponding variation of construction costs is asymmetric. When E increases by 50% (to 75 MPa), the construction cost decreases by less than 20%. In contrast, when E decreases by 50% (to 25 MPa), the construction cost increases by more than 100%.

Fig. 4 shows the variation of construction cost as a function of

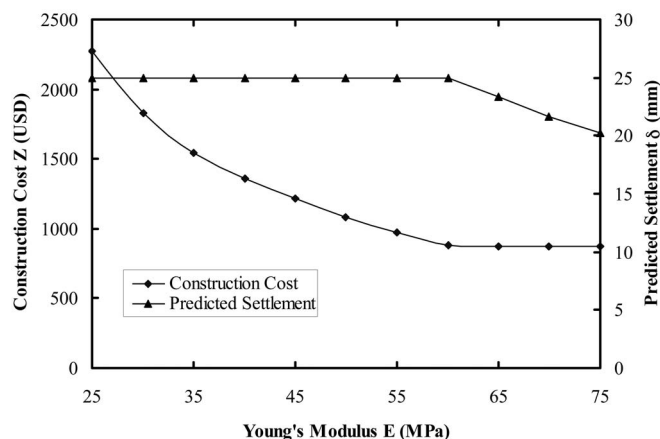


Fig. 4. Effect of Young's modulus E

E . The cost decreases nonlinearly, as E increases from 25 to 60 MPa, and it remains constant thereafter. Since E is related to the design requirement of allowable settlement [Eq. (9)], Fig. 4 also shows the calculated settlement in the economically optimized design by solid triangles. When E is less than 60 MPa, the calculated allowable settlement is 25 mm, which equals δ_r , and indicates that δ_r controls the final design. In this case, change of E leads to change of the final design. Additional efforts might be warranted for improving the characterization of E . On the other hand, when E is larger than 60 MPa, the calculated settlement is less than 25 mm. This case indicates that δ_r does not control the final design, and therefore the δ_r requirement is automatically satisfied by meeting the FS_r requirement (refer to the next section for details). Accordingly, change of E has no effect on the final design, and subsequently no additional efforts are warranted to determine the precise value of E .

The second sensitivity study was performed with a variation of $\pm 10\%$ of their baseline values for all four soil parameters. Fig. 3(b) shows the variations of construction costs with the baseline construction cost. The figure also shows that E and ϕ' are the two parameters that significantly affect the construction costs and deserve more attention in site investigations. The variations of construction costs with soil properties enable a quantitative cost-benefit analysis that can be performed to justify whether additional investment is warranted for improving soil property characterization.

Effect of Design Requirements

Design requirements are paramount, and their variation can result in different designs and consequently different construction costs. Because an economically optimized design incorporates construction cost estimates, it is possible to explore the effect of design requirements on the construction costs. Characterization of the design requirement effect is particularly valuable in performance-based design (FEMA 1996; Ghobarah 2001), in which facility owners make the determination, with their engineers, of the desired performance level of their facility (i.e., design requirements). Construction costs are one key concern.

Fig. 5 shows the variation of construction costs as functions of both FS_r and δ_r . For a given δ_r , the construction costs increase as FS_r increases from 1.5 to 5.0. FS_r affects the economically optimized design and therefore the construction cost. When $FS_r = 1.5$, the construction costs decrease significantly, with a decrease-

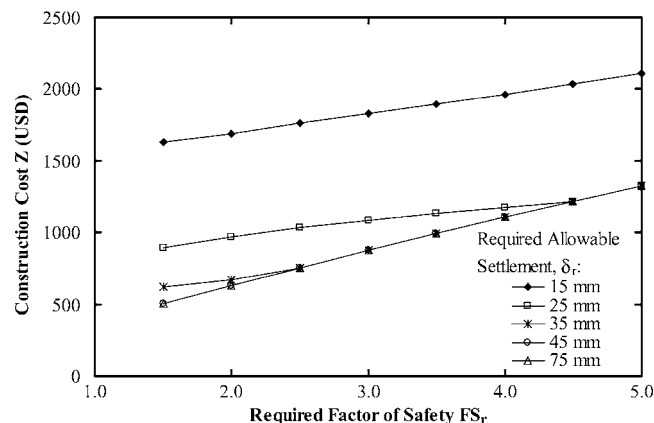


Fig. 5. Effects of design requirements FS_r and δ_r

ing rate as δ_r increases from 15 to 45 mm, and they appear to remain constant when $\delta_r > 45$ mm. When δ_r is relatively small, the effect of δ_r is significant on the design and construction costs. However, the effect decreases as δ_r increases, and it becomes negligible when δ_r is larger than a threshold value (e.g., 45 mm for $FS_r = 1.5$). The threshold value decreases from 45 to 25 mm as FS_r increases from 1.5 to 4.5. When δ_r is larger than the threshold value, it has no effect on the design, and therefore it is not a governing design requirement. The increase of the threshold value implies that the effect of δ_r diminishes as FS_r increases. This variation suggests that, in spread footing design, a relatively stringent ULS requirement (i.e., large FS_r) generally ensures fulfillment of the SLS requirement (i.e., δ_r).

Summary and Conclusions

In this paper, a design approach was developed that explicitly considers construction economics and results in a geo-structure with the minimum construction cost. The approach is expressed as an optimization process, in which the optimization objective is to minimize construction cost. Design parameters, such as the dimensions of the geo-structures, are treated as optimization variables, which vary within the ranges constrained by the design requirements, such as ULS and SLS requirements. As the optimization is solved, a set of design parameters is obtained, and the geo-structure is specified or designed that not only satisfies all design requirements, but also results in the minimum construction cost.

One key component in the economically optimized design is the construction cost estimate, which consists of quantity takeoff and cost estimates using unit prices. A simplified procedure is developed to acquire the unit prices needed in the cost estimates, and the procedure is illustrated for an example spread footing. Unit prices are developed for the five construction work tasks, including excavation, formwork, reinforcement, concrete, and compacted backfill.

Comparison of the economically optimized design example with conventional designs shows that the savings in construction cost could be as much as 30%. Furthermore, because the unit prices and construction costs are geographically dependent, the economically optimized design can vary by locale. For the same ground conditions and design requirements, footing designs for the United States national average are different from those in New York City or Beijing. The design in Beijing makes use of the

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Finally, it should be noted that the economically optimized design framework is independent of the calculation models used for q_{ult} and δ and the types of structures under design.

The work described in this paper was supported by grants from City University of Hong Kong (Project Nos. 7002072 and 7200070).

Fig. 6 shows the optimization model for spread footing design in a Microsoft Excel spreadsheet. The spreadsheet is divided into three zones: (1) an input data zone from Rows 1 to 7; (2) a design parameter and calculation zone from Rows 9 to 18; and (3) a cost estimate zone starting from Row 20.

The optimization model then is set up using the Excel toolkit Solver, which can be launched by going to the drop-down menu of “Tools” and selecting “Solver.” As shown in the lower right corner of Fig. 6, the Solver parameters window appears that permits definition of the optimization objective, variables, and constraints. The optimization objective is defined by assigning Cell D22 as the target cell and checking the “Min” button. Cells B10–B12 are the optimization variables. Both design requirements and practical constraints are treated as constraints and implemented as Cell B10 ≤ 2 , Cell B10 ≥ 0.5 , Cell B11 ≥ 0 , Cell B12 ≥ 0 , Cell



E18 $\geq$ Cell H3, and Cell H12 $\leq$ Cell H4. The optimization is executed by clicking the “Solve” button in the upper right corner, and the minimum construction cost and the corresponding design parameters and calculated FS and δ are generated. For the example shown in Fig. 6, the minimum construction cost is USD 1,086.

Note that the Solver toolkit may not have been installed by default, but it can be activated by going to the drop-down menu of “Tools,” selecting “Add-Ins . . .,” and checking the “Solver Add-In” checkbox.

Notation

The following symbols are used in this paper:

B	= footing width;
B_0	= over-excavation distance along width direction;
c_b, c_c, c_e, c_f, c_r	= unit prices of compacted backfill, concrete, excavation, formwork, and reinforcement, respectively;
$c_1, c_2, c_3, \dots, c_n$	= unit costs of construction materials, labor, and so on;
D	= footing depth;
D_l	= lower bound of footing depth;
D_u	= upper bound of footing depth;
E	= Young's modulus;
F	= vertical load;
FS	= factor of safety;
FS_r	= Required factor of safety;
H	= footing height;
k	= proportionality coefficient in Eq. (13);
L	= footing length;
L_0	= overexcavation distance along length direction;
N_γ, N_q	= bearing capacity factors;
$PI_1, PI_2, PI_3, \dots, PI_n$	= performance indexes;
$PI_{1r}, PI_{2r}, PI_{3r}, \dots, PI_{nr}$	= required performance indexes;
Q_b, Q_c, Q_e, Q_f, Q_r	= quantities of compacted backfill, concrete, excavation, formwork, and reinforcement, respectively;
q'	= effective overburden stress at foundation level;
q_{ult}	= ultimate bearing capacity;
x_{il}	= lower bound of design parameter x_i ;
x_{iu}	= upper bound of design parameter x_i ;
$x_1, x_2, x_3, \dots, x_n$	= design parameters;
Z	= construction cost;
β_z	= shape factor for settlement calculation given by Fig. 2;
γ	= soil unit weight;

γ'	= effective unit weight of soil below footing;
δ	= calculated settlement;
δ_r	= required allowable settlement;
$\zeta_{\gamma d}, \zeta_{qd}$	= depth factors in bearing capacity equation;
$\zeta_{\gamma s}, \zeta_{qs}$	= shape factors in bearing capacity equation;
ν	= Poisson's ratio; and
ϕ'	= effective friction angle.

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